Neuro OSCE Prep — Cases 3 & 4

Multiple Sclerosis · Physiotherapy Clinical Reasoning Guide

Target: 70%+ (Very Good)

Case 3 — Ms. Sophie Bennett (RRMS, 34F)

Case 4 — Mrs. Catherine Holt (SPMS, 52F)

CASE 3

Ms. Sophie Bennett — 34F · RRMS

Post-relapse (optic neuritis) · Fatigue · Leg heaviness >200 m · Bilateral foot paraesthesia · Balance impairment when fatigued · Fear of exercise-triggered relapse

1 Subjective Assessment ▲

CONSENT & OPENING STATEMENT

"Good morning Ms. Bennett. My name is [Name] and I am the physiotherapist seeing you today. Before we begin, I want to explain that I'll ask you some questions about your MS and how it's been affecting you, and then do some physical assessments. Everything you share is confidential and is used only to plan the best care for you. Please feel free to stop or rest at any point. Are you happy to proceed?"

KEY QUESTIONS + CLINICAL REASONING

1. Tell me about your fatigue — when is it worst, what triggers it, and does it improve with rest?

> 💡 MS fatigue is multifactorial. I need to distinguish **primary central fatigue** (driven by increased cortical effort to conduct signals along demyelinated axons) from **secondary fatigue** (deconditioning, disturbed sleep, depression). I also want to identify Uhthoff's phenomenon — symptom worsening with heat — which directly shapes my exercise environment and scheduling. This question drives my fatigue management strategy.

2. Can you describe the leg heaviness when you walk — at what point does it come on, does it affect both legs equally, and what does the walking look like when it gets bad?

> 💡 This characterises the functional impairment in detail. I want to know: is this weakness, spasticity, fatigue, or foot drop? Does it follow a consistent threshold (suggesting heat/metabolic loading of demyelinated pathways) or is it variable? The answer determines whether I target strength, gait re-education, ankle-foot orthosis referral, or energy conservation first.

3. Describe the pins and needles in your feet — constant or intermittent, does it affect your confidence or balance on different surfaces?

> 💡 Sensory loss in the feet reduces proprioceptive afferent input to the cerebellum and somatosensory cortex. This directly degrades postural stability. I need to know whether Sophie's balance relies heavily on visual input to compensate (suggesting dorsal column involvement), which shapes my balance retraining approach. It also informs whether sensory substitution strategies are appropriate.

4. Have you had any falls or near-falls in the last three months — and does your balance feel different in the dark, or on uneven ground?

> 💡 Fall history is the strongest single predictor of future falls in MS. This determines the urgency and intensity of my balance programme. Asking

about darkness and uneven terrain probes whether she's compensating with vision — if she's much worse in the dark, that confirms proprioceptive impairment is dominant over vestibular. This directly changes my exercise design.

5. You mentioned being terrified exercise might cause a relapse — can you tell me more about where that belief came from?

> 💡 This is potentially the most important question I ask. Fear-avoidance and kinesiophobia are major barriers to rehabilitation in RRMS. If I don't address this belief explicitly and with evidence, she will not adhere to any programme I prescribe. Understanding the source of the fear tells me whether to use brief education, motivational interviewing, or a graded exposure approach. The evidence is clear: moderate exercise does not trigger relapses in RRMS.

6. How is your vision since the optic neuritis — any residual blurring, colour change, or depth perception problems?

> 💡 Residual visual deficits from optic neuritis impair the visual compensation that her nervous system is likely using to offset the proprioceptive loss. If vision is compromised, her balance risk is higher than it appears — and I cannot rely on visual feedback-based balance exercises as heavily. I need to know this before designing her balance programme.

RED FLAGS

🚨 **New or worsening bladder / bowel dysfunction**

Significance: May indicate a new relapse with spinal cord involvement (transverse myelitis pattern). Also a major fall risk — urgency incontinence causes rushed, unbalanced movement to the toilet. Can precipitate significant distress.

Action: Urgent MS nurse / neurologist notification. Continence physiotherapy referral. Withhold intensive exercise until relapse ruled out.

🚨 New neurological deficit within the past 4–6 weeks (possible active relapse)

Significance: A relapse is defined as new or worsening neurological symptoms lasting >24 hours in the absence of fever/infection. Exercise during an active relapse can worsen symptoms and delays recovery.

Action: Refer back to neurologist urgently. Suspend formal exercise programme until relapse confirmed resolved (typically ≥4 weeks post-relapse onset).

🚨 Lhermitte's sign (new onset)

Significance: An electric shock sensation down the spine on neck flexion — indicates active cervical cord demyelination. New onset suggests an evolving lesion or early relapse.

Action: Avoid all cervical flexion-loaded activities. Urgent neurologist referral. MRI cervical spine may be indicated.

🚨 New or rapidly worsening visual loss

Significance: Given the recent optic neuritis, any new visual loss could represent a second attack on the optic nerve. Bilateral optic neuritis should raise suspicion of NMOSD rather than RRMS.

Action: Do not proceed with assessment. Urgent ophthalmology and neurology referral. Consider aquaporin-4 antibody serology.

🚨 Severe or persistent depression / suicidal ideation

Significance: Depression prevalence in MS is approximately 50%. Severe depression makes meaningful engagement with rehabilitation impossible and carries significant personal risk.

Action: PHQ-9 screening. Urgent GP or psychology referral. Involve MS specialist nurse. Do not prioritise exercise over mental health crisis.

PROBLEM LIST

Domain	Problems Identified

Physical Impairments	• Primary central MS fatigue • Bilateral lower limb sensory impairment (paraesthesia — feet) • Balance impairment, worse with fatigue and heat • Functional lower limb weakness / leg heaviness on exertion • Likely residual visual deficit from optic neuritis
Functional Activity Limitations	• Walking limited to <200 m before symptom onset • Reduced confidence and safety with community ambulation • Fall risk (balance compromised under fatigue loading)
Participation Restrictions	• Reduced community participation (outings, social activities) • Potential impact on employment and daily roles • Avoidance of physical activity due to relapse fear

PSYCHOSOCIAL / YELLOW FLAGS

⚠ Kinesiophobia / Fear-Avoidance Behaviour

Sophie describes being "terrified" that exercise will trigger a relapse. This is a well-documented maladaptive belief in RRMS that strongly predicts exercise avoidance and deconditioning. Without early, targeted education and confidence-building, her physical condition will continue to decline regardless of what I prescribe.

⚠ Anxiety about Disease Progression

A recent relapse commonly heightens existential fear about the disease trajectory. I will validate her concerns, use motivational interviewing principles, and co-produce goal-setting to rebuild her sense of agency. Psychology or MS counselling referral if anxiety is severe.

2 **Pathophysiology of Working Diagnosis** ▼

3 **Differential Diagnoses** ▼

4 **Objective Assessment & Special Tests** ▼

5 **Treatment Plan** ▼

CASE 4

Mrs. Catherine Holt — 52F · SPMS

2-year progressive decline · Rolling frame (all indoor mobility) · Bilateral adductor spasticity · Personal care and transfers impaired · Lhermitte's sign

1 **Subjective Assessment** ▲

CONSENT & OPENING STATEMENT

"Good morning Mrs. Holt. My name is [Name] and I'm your physiotherapist today. I'd like to start by hearing from you about how things have been — then I'll do some physical assessments to understand your movement and muscle function in more detail. Please stop me or take a break at any point;

there's no rush. Do you have any questions before we start? Are you happy to proceed?"

> 💡 With SPMS, I signal from the opening that I respect her pace, understand that fatigue and effort cost are real, and that I'm there to work with her priorities — not impose a rehabilitation agenda she hasn't bought into.

KEY QUESTIONS + CLINICAL REASONING

1. Can you describe the spasticity in your legs — when is it at its worst, what triggers it, and how does it interfere with your personal care and transfers?

> 💡 Spasticity in SPMS is the central clinical problem and is the prerequisite barrier to everything else. I need to understand: is it worse in the morning (diurnal pattern — common in UMN spasticity), triggered by specific movements, or constantly severe? Identifying triggers is critical because the most effective intervention may be treating a precipitating cause (UTI, constipation, pressure area) rather than physiotherapy alone. The impact on personal care and transfers defines my functional goal-setting.

2. Have you had any changes in your bladder or bowel recently — urgency, frequency, incomplete emptying, or constipation?

> 💡 This is simultaneously a red flag screen and a spasticity precipitant inquiry — the two most important functions of the same question. Bladder dysfunction affects 75–80% of people with MS. A urinary tract infection or bladder distension is the single most common cause of sudden spasticity increase. If present, treating the UTI may resolve the spasticity escalation without any additional physiotherapy intervention. I must ask this before I do anything else.

3. Tell me about Lhermitte's sign — is it longstanding or has it changed recently, what position triggers it, and how long does the sensation last?

> 💡 Lhermitte's sign indicates cervical dorsal column pathology. Longstanding, stable Lhermitte's tells me to avoid cervical flexion-loaded positions in treatment — relevant to positioning, exercise, and hydrotherapy. New or worsening Lhermitte's is a potential red flag suggesting active disease. The answer directly restricts or permits certain physiotherapy techniques and positions.

4. Walk me through a typical transfer — for example, bed to chair and chair to toilet. At exactly which point does it break down, and what does it look like when it goes wrong?

> 💡 Transfer safety is Catherine's highest-priority functional problem. I need a task-analysis-level understanding of the transfer sequence to identify the specific failure point: is it the initial rise from sitting (quadriceps/hip extensor weakness), the pivot step (adductor spasticity closing the legs), or the controlled lowering (eccentric control failure)? Identifying the exact failure point allows me to prescribe a targeted intervention rather than a generic transfer training programme.

5. Are you having any pain — particularly in the legs, hips, or lower back — and have you or your carer noticed any skin changes, redness, or pressure areas?

> 💡 Pain in SPMS is commonly neuropathic (dysaesthesia from dorsal horn sensitisation) or musculoskeletal secondary to spasticity (adductor-origin groin pain, hip flexor tightness). Both worsen spasticity via nociceptive input to the spinal cord. Skin breakdown is both a medical emergency and a major spasticity precipitant. I must screen proactively — patients with limited mobility often do not spontaneously report pressure changes.

6. Who helps you with your care at home, and how are they managing with the transfers? Are there any concerns about their safety or yours during care?

> 💡 Carer capacity and technique directly determine whether my rehabilitation plan is deliverable at home. If the carer is using unsafe

manual handling, both Catherine and the carer are at injury risk. I need to know this so I can include carer moving-and-handling training in my plan. Carer burden and burnout are also real — if the carer is struggling, the whole care package is at risk of breakdown.

RED FLAGS

🚨 Sudden or significantly worsening Lhermitte's sign

Significance: May indicate a new active cervical cord lesion, cord compression (rare but possible — atlantoaxial or subaxial), or evolving myelopathy. New or dramatically worsened Lhermitte's is a neurological review trigger.

Action: Avoid all cervical flexion-loaded activities immediately. Urgent neurologist referral. MRI cervical spine may be required. Suspend neck-involving physiotherapy until reviewed.

🚨 Rapidly escalating spasticity over days or weeks

Significance: In SPMS, sudden spasticity escalation is almost always triggered by a noxious stimulus — UTI, pressure ulcer, deep vein thrombosis, ingrowing toenail, bowel impaction, fracture — rather than disease progression alone. Missing this keeps the precipitant untreated and makes all physiotherapy interventions ineffective.

Action: Systematic screen for all noxious precipitants. Urgent GP referral for UTI/urinalysis. Do not escalate physiotherapy intensity until the cause is identified and treated.

🚨 New or worsening pressure ulcer / skin breakdown

Significance: In SPMS with reduced mobility and sustained spasticity, pressure ulcers are a major complication. They cause significant pain, are a potent spasticity trigger, and can lead to life-threatening sepsis.

Action: Urgent tissue viability nurse referral. Immediate seating and pressure relief review. Halt exercise over or adjacent to affected areas. Document and escalate same day.

🚨 Autonomic dysreflexia features (if high spinal cord involvement suspected)

Significance: In MS with high thoracic or cervical cord lesions, noxious stimuli can trigger a sudden, massive, potentially fatal sympathetic surge: severe hypertension, pounding headache, facial flushing, profuse sweating, bradycardia. A medical emergency.

Action: Immediately sit patient upright; identify and remove the trigger (full bladder, tight clothing, pressure). Call for emergency medical help. Do not continue treatment.

🚨 Signs of deep vein thrombosis in lower limb

Significance: Reduced mobility in SPMS significantly elevates DVT risk. Unilateral leg swelling, warmth, redness, and calf tenderness warrant urgent investigation. A DVT is also a major spasticity precipitant.

Action: Wells Score assessment; urgent GP/ED referral for Doppler ultrasound. Withhold passive stretching of the affected limb until excluded.

PROBLEM LIST

Domain	Problems Identified
Physical Impairments	• Bilateral lower limb spasticity — adductors predominant (UMN pattern, progressive) • Significant bilateral lower limb weakness (rolling frame required for all indoor mobility) • Cervical cord vulnerability — Lhermitte's sign, restricts treatment positions • Likely pain secondary to spasticity (groin, hip, adductor origin) • Likely bladder / bowel dysfunction (spasticity precipitant + independent impairment)
Functional Activity Limitations	• All transfers (bed, chair, toilet) compromised by spasticity and weakness

	• Personal care limited — adductor spasticity closes legs, impeding perineal access • All indoor mobility requires rolling frame; outdoor mobility likely absent or minimal • Dependence on carer for most ADLs
Participation Restrictions	• Loss of independence — significant impact on identity and dignity • Social isolation risk • Carer burden — transfers may be unsafe with current technique • High likelihood of employment cessation (age 52, 2-year progressive decline)

PSYCHOSOCIAL / YELLOW FLAGS

⚠ Grief and Adjustment to Progressive Disability

Secondary progressive MS is a diagnosis of unremitting decline. Catherine will be navigating loss of independence, roles, and identity. I use a "What Matters to You" framework for goal-setting — focusing on her priorities (dignity in personal care, safe transfers) rather than imposing impairment-based goals. Psychology or MS counselling referral if adjustment difficulties are significant.

⚠ Carer Burden and Fatigue

If the carer is struggling — physically, emotionally, or practically — the care package is at risk of breakdown, and both parties are at injury risk from unsafe manual handling. I will involve the carer explicitly in all sessions, provide formal moving-and-handling training, and escalate carer support needs to social work and community care services if required.

2 Pathophysiology of Working Diagnosis ▼

3 **Differential Diagnoses** ▼

4 **Objective Assessment & Special Tests** ▼

5 **Treatment Plan** ▼